

OpenStax Calculus Volume 3

Section 5.1 Exercises: Double Integrals over Rectangular Regions

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In the following exercises, use the midpoint rule with $m = 4$ and $n = 2$ to estimate the volume of the solid bounded by the surface $z = f(x, y)$, the vertical planes $x = 1$, $x = 2$, $y = 1$, and $y = 2$, and the horizontal plane $z = 0$.

1. $f(x, y) = 4x + 2y + 8xy$
2. $f(x, y) = 16x^2 + \frac{y}{2}$

In the following exercises, estimate $\iint_R f(x, y) dA$ by using a Riemann sum with $m = n = 2$ and the sample points to be the lower left corners of the subrectangles of the partition.

3. $f(x, y) = \sin x - \cos y$, $R = [0, \pi] \times [0, \pi]$
4. $f(x, y) = \cos x + \cos y$, $R = [0, \pi] \times [0, \frac{\pi}{2}]$
5. Use the midpoint rule with $m = n = 2$ to estimate $\iint_R f(x, y) dA$, where the values of the function f on $R = [8, 10] \times [9, 11]$ are given in the following table.

	y				
x	9	9.5	10	10.5	11
8	9.8	5	6.7	5	5.6
8.5	9.4	4.5	8	5.4	3.4
9	8.7	4.6	6	5.5	3.4
9.5	6.7	6	4.5	5.4	6.7
10	6.8	6.4	5.5	5.7	6.8

6. The values of the function f on the rectangle $R = [0, 2] \times [7, 9]$ are given in the following table. Estimate the double integral $\iint_R f(x, y) dA$ by using a Riemann sum with $m = n = 2$. Select the sample points to be the upper right corners of the subsquares of R .

	$y_0 = 7$	$y_1 = 8$	$y_2 = 9$
$x_0 = 0$	10.22	10.21	9.85
$x_1 = 1$	6.73	9.75	9.63
$x_2 = 2$	5.62	7.83	8.21

7. The depth of a children's 4-ft by 4-ft swimming pool, measured at 1-ft intervals, is given in the following table.

- (a) Estimate the volume of water in the swimming pool by using a Riemann sum with $m = n = 2$. Select the sample points using the midpoint rule on $R = [0, 4] \times [0, 4]$.
- (b) Approximate the average depth of the swimming pool.

x	y				
	0	1	2	3	4
0	1	1.5	2	2.5	3
1	1	1.5	2	2.5	3
2	1	1.5	1.5	2.5	3
3	1	1	1.5	2	2.5
4	1	1	1	1.5	2

8. The depth of a 3-ft by 3-ft hole in the ground, measured at 1-ft intervals, is given in the following table.

- (a) Estimate the volume of the hole by using a Riemann sum with $m = n = 3$ and the sample points to be the upper left corners of the subsquares of R .
- (b) Approximate the average depth of the hole.

x	y			
	0	1	2	3
0	6	6.5	6.4	6
1	6.5	7	7.5	6.5
2	6.5	6.7	6.5	6
3	6	6.5	5	5.6

9. The level curves $f(x, y) = k$ of the function f are given in the following graph, where k is a constant.

- (a) Apply the midpoint rule with $m = n = 2$ to estimate the double integral $\iint_R f(x, y) dA$, where $R = [0.2, 1] \times [0, 0.8]$.

- (b) Estimate the average value of the function f on R .

[Figure: A series of curves marked $k = -1, -\frac{1}{2}, -\frac{1}{4}, -\frac{1}{8}, 0, \frac{1}{8}, \frac{1}{4}, \frac{1}{2},$ and 1 . The line marked $k = 0$ serves as an asymptote along the line $y = x$. The lines originate at $1, 0.7, 0.5, 0.38,$ and 0 along the y -axis and at $0.38, 0.5, 0.7,$ and 1 along the x -axis, with the further-out lines curving less dramatically toward the asymptote.]

10. The level curves $f(x, y) = k$ of the function f are given in the following graph, where k is a constant.

- (a) Apply the midpoint rule with $m = n = 2$ to estimate the double integral $\iint_R f(x, y) dA$, where $R = [0.1, 0.5] \times [0.1, 0.5]$.

- (b) Estimate the average value of the function f on R .

[Figure: A series of quarter circles drawn in the first quadrant marked $k = \frac{1}{32}, \frac{1}{16}, \frac{1}{8},$

$\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, and 1. The quarter circles have radii 0.17, 0.25, 0.35, 0.5, 0.71, 0.87, and 1, respectively.]

11. The solid lying under the surface $z = \sqrt{4 - y^2}$ and above the rectangular region $R = [0, 2] \times [0, 2]$ is illustrated in the following graph. Evaluate the double integral $\iint_R f(x, y) dA$, where $f(x, y) = \sqrt{4 - y^2}$, by finding the volume of the corresponding solid.
[Figure: A quarter cylinder with center along the x -axis and with radius 2. It has height 2 as shown.]
12. The solid lying under the plane $z = y + 4$ and above the rectangular region $R = [0, 2] \times [0, 4]$ is illustrated in the following graph. Evaluate the double integral $\iint_R f(x, y) dA$, where $f(x, y) = y + 4$, by finding the volume of the corresponding solid.
[Figure: In xyz -space, a shape is created with sides given by $y = 0$, $x = 0$, $y = 4$, $x = 2$, $z = 0$, and the plane that runs from $z = 4$ along the y -axis to $z = 8$ along the plane formed by $y = 4$.]

In the following exercises, calculate the integrals by interchanging the order of integration.

13. $\int_{-1}^1 (\int_{-2}^2 (2x + 3y + 5) dx) dy$
14. $\int_0^2 (\int_0^1 (x + 2e^y - 3) dx) dy$
15. $\int_1^{27} (\int_1^2 (\sqrt[3]{x} + \sqrt[3]{y}) dy) dx$
16. $\int_1^{16} (\int_1^8 (\sqrt[4]{x} + 2\sqrt[3]{y}) dy) dx$
17. $\int_{\ln 2}^{\ln 3} (\int_0^l e^{x+y} dy) dx$
18. $\int_0^2 (\int_0^1 3^{x+y} dy) dx$
19. $\int_1^6 (\int_2^9 \frac{\sqrt{y}}{x^2} dy) dx$
20. $\int_1^9 (\int_4^2 \frac{\sqrt{x}}{y^2} dy) dx$

In the following exercises, evaluate the iterated integrals by choosing the order of integration.

21. $\int_0^\pi \int_0^{\pi/2} \sin(2x) \cos(3y) dx dy$
22. $\int_{\pi/12}^{\pi/8} \int_{\pi/4}^{\pi/3} [\cot x + \tan(2y)] dx dy$
23. $\int_1^e \int_1^e [\frac{1}{x} \sin(\ln x) + \frac{1}{y} \cos(\ln y)] dx dy$
24. $\int_1^e \int_1^e \frac{\sin(\ln x) \cos(\ln y)}{xy} dx dy$
25. $\int_1^2 \int_1^2 (\frac{\ln y}{x} + \frac{x}{2y+1}) dy dx$
26. $\int_1^e \int_1^2 x^2 \ln(x) dy dx$
27. $\int_1^{\sqrt{3}} \int_1^2 y \arctan(\frac{1}{x}) dy dx$
28. $\int_0^1 \int_0^{1/2} (\arcsin x + \arcsin y) dy dx$
29. $\int_0^1 \int_1^2 x e^{x+4y} dy dx$

30. $\int_1^2 \int_0^1 x e^{x-y} dy dx$
31. $\int_1^e \int_1^e (\frac{\ln y}{\sqrt{y}} + \frac{\ln x}{\sqrt{x}}) dy dx$
32. $\int_1^e \int_1^e (\frac{x \ln y}{\sqrt{y}} + \frac{y \ln x}{\sqrt{x}}) dy dx$
33. $\int_0^1 \int_1^2 (\frac{x}{x^2+y^2}) dy dx$
34. $\int_0^1 \int_1^2 \frac{y}{x+y^2} dy dx$

In the following exercises, find the average value of the function over the given rectangles.

35. $f(x, y) = -x + 2y$, $R = [0, 1] \times [0, 1]$
36. $f(x, y) = x^4 + 2y^3$, $R = [1, 2] \times [2, 3]$
37. $f(x, y) = \sinh x + \sinh y$, $R = [0, 1] \times [0, 2]$
38. $f(x, y) = \arctan(xy)$, $R = [0, 1] \times [0, 1]$
39. Let f and g be two continuous functions such that $0 \leq m_1 \leq f(x) \leq M_1$ for any $x \in [a, b]$ and $0 \leq m_2 \leq g(y) \leq M_2$ for any $y \in [c, d]$. Show that the following inequality is true:
 $m_1 m_2 (b-a)(d-c) \leq \int_a^b \int_c^d f(x)g(y) dy dx \leq M_1 M_2 (b-a)(d-c)$.

In the following exercises, use property v. of double integrals and the answer from the preceding exercise to show that the following inequalities are true.

40. $\frac{1}{e^2} \leq \iint_R e^{-x^2-y^2} dA \leq 1$, where $R = [0, 1] \times [0, 1]$
41. $\frac{\pi^2}{144} \leq \iint_R \sin x \cos y dA \leq \frac{\pi^2}{48}$, where $R = [\frac{\pi}{6}, \frac{\pi}{3}] \times [\frac{\pi}{6}, \frac{\pi}{3}]$
42. $0 \leq \iint_R e^{-y} \cos x dA \leq (\frac{\pi}{2})^2$, where $R = [0, \frac{\pi}{2}] \times [0, \frac{\pi}{2}]$
43. $0 \leq \iint_R (\ln x)(\ln y) dA \leq (e-1)^2$, where $R = [1, e] \times [1, e]$
44. Let f and g be two continuous functions such that $0 \leq m_1 \leq f(x) \leq M_1$ for any $x \in [a, b]$ and $0 \leq m_2 \leq g(y) \leq M_2$ for any $y \in [c, d]$. Show that the following inequality is true:
 $(m_1 + m_2)(b-a)(d-c) \leq \int_a^b \int_c^d [f(x) + g(y)] dy dx \leq (M_1 + M_2)(b-a)(d-c)$.

In the following exercises, use property v. of double integrals and the answer from the preceding exercise to show that the following inequalities are true.

45. $\frac{2}{e} \leq \iint_R (e^{-x^2} + e^{-y^2}) dA \leq 2$, where $R = [0, 1] \times [0, 1]$
46. $\frac{\pi^2}{36} \leq \iint_R (\sin x + \cos y) dA \leq \frac{\pi^2 \sqrt{3}}{36}$, where $R = [\frac{\pi}{6}, \frac{\pi}{3}] \times [\frac{\pi}{6}, \frac{\pi}{3}]$
47. $\frac{\pi^2}{4} e^{-\frac{\pi}{2}} \leq \iint_R (\cos x + e^{-y}) dA \leq \frac{\pi^2}{2}$, where $R = [0, \frac{\pi}{2}] \times [0, \frac{\pi}{2}]$
48. $0 \leq \iint_R (\ln x + \ln y) dA \leq 2(e-1)^2$, where $R = [1, e] \times [1, e]$

In the following exercises, the function f is given in terms of double integrals.

- (a) Determine the explicit form of the function f .
- (b) Find the volume of the solid under the surface $z = f(x, y)$ and above the region R .
- (c) Find the average value of the function f on R .
- (d) Use a computer algebra system (CAS) to plot $z = f(x, y)$ and $z = f_{\text{ave}}$ in the same system of coordinates.

49. [T] $f(x, y) = \int_0^y \int_0^x (xs + yt) ds dt$, where $(x, y) \in R = [0, 1] \times [0, 1]$
50. [T] $f(x, y) = \int_0^x \int_0^y [\cos(s) + \cos(t)] dt ds$, where $(x, y) \in R = [0, 3] \times [0, 3]$
51. Show that if f and g are continuous on $[a, b]$ and $[c, d]$, respectively, then $\int_a^b \int_c^d [f(x) + g(y)] dy dx = (d - c) \int_a^b f(x) dx + \int_a^b \int_c^d g(y) dy dx = (b - a) \int_c^d g(y) dy + \int_c^d \int_a^b f(x) dx dy$.
52. Show that $\int_a^b \int_c^d [yf(x) + xg(y)] dy dx = \frac{1}{2}(d^2 - c^2) \left(\int_a^b f(x) dx \right) + \frac{1}{2}(b^2 - a^2) \left(\int_c^d g(y) dy \right)$.
53. [T] Consider the function $f(x, y) = e^{-x^2-y^2}$, where $(x, y) \in R = [-1, 1] \times [-1, 1]$.
 - (a) Use the midpoint rule with $m = n = 2, 4, \dots, 10$ to estimate the double integral $I = \iint_R e^{-x^2-y^2} dA$. Round your answers to the nearest hundredths.
 - (b) For $m = n = 2$, find the average value of f over the region R . Round your answer to the nearest hundredths.
 - (c) Use a CAS to graph in the same coordinate system the solid whose volume is given by $\iint_R e^{-x^2-y^2} dA$ and the plane $z = f_{\text{ave}}$.
54. [T] Consider the function $f(x, y) = \sin(x^2) \cos(y^2)$, where $(x, y) \in R = [-1, 1] \times [-1, 1]$.
 - (a) Use the midpoint rule with $m = n = 2, 4, \dots, 10$ to estimate the double integral $I = \iint_R \sin(x^2) \cos(y^2) dA$. Round your answers to the nearest hundredths.
 - (b) For $m = n = 2$, find the average value of f over the region R . Round your answer to the nearest hundredths.
 - (c) Use a CAS to graph in the same coordinate system the solid whose volume is given by $\iint_R \sin(x^2) \cos(y^2) dA$ and the plane $z = f_{\text{ave}}$.

In the following exercises, the functions f_n are given, where $n \geq 1$ is a natural number.

- (a) Find the volume of the solids S_n under the surfaces $z = f_n(x, y)$ and above the region R .
- (b) Determine the limit of the volumes of the solids S_n as n increases without bound.

55. $f_n(x, y) = x^n + y^n + xy$, $(x, y) \in R = [0, 1] \times [0, 1]$
56. $f_n(x, y) = \frac{1}{x^n} + \frac{1}{y^n}$, $(x, y) \in R = [1, 2] \times [1, 2]$
57. Show that the average value of a function f on a rectangular region $R = [a, b] \times [c, d]$ is $f_{\text{ave}} \approx \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*)$, where (x_{ij}^*, y_{ij}^*) are the sample points of the partition of R , where $1 \leq i \leq m$ and $1 \leq j \leq n$.

58. Use the midpoint rule with $m = n$ to show that the average value of a function f on a rectangular region $R = [a, b] \times [c, d]$ is approximated by

$$f_{\text{ave}} \approx \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n f\left(\frac{x_{i-1} + x_i}{2}, \frac{y_{j-1} + y_j}{2}\right).$$

59. An isotherm map is a chart connecting points having the same temperature at a given time for a given period of time. Use the preceding exercise and apply the midpoint rule with $m = n = 2$ to find the average temperature over the region given in the following figure.

[Figure: A contour map showing surface temperature in degrees Fahrenheit. Given the map, the midpoint rule would give rectangles with values 71, 72, 40, and 43.]